Available online at www.ijpe-online.com

International Journal of Performability Engineering

vol. 20, no. 3, March 2024, pp. 149-156
DOI: 10.23940/ijpe.24.03.p3.149156

Reliability Assessment of Distribution System Grid-Connected Multi-Inverter for Solar Photo-Voltaic Systems: A Case Study

Darius Muyizere[a,*], Arcade Nshimiyimana[b], Theophile Mugerwa[c], Lawrence K. Letting[d], and Bernard B. Munyazikwiye[e]

[a]*African Centre of Excellence in Energy and Sustainable Development, University of Rwanda, Kigali, Rwanda*
[b]*Department of Electrical and Electronics Engineering, IPRC Tumba - Rwanda Polytechnic, Northern Province, Rwanda*
[c]*Department of Geology, University of Rwanda, Kigali, Rwanda*
[d]*Electrical and Communications Engineering, Moi University, Eldoret, Kenya*
[e]*Department of Mechanical and Energy Engineering, University of Rwanda, Kigali, Rwanda*

Abstract

The contemporary electrical grid aims to deliver safe and reliable energy to consumers. As technology advances and distributed generation (DG) becomes more prevalent, the distribution system grows increasingly complex and decentralized. The swift integration of renewable energy sources, driven by concerns about global warming and carbon emissions, adds to this complexity. Given the direct link between the distribution system and consumer needs, reliability is paramount. However, the current distribution system faces operational issues, necessitating a dependable, resilient power system devoid of interruptions and glitches. Distributed generation (DG), along with its grid integration, holds promise in significantly enhancing the reliability of the existing distribution system. This research studied a case study of distribution system reliability evaluation using solar PV in the system. Various instances were addressed, and we found that after implementing DG, system dependability increased. For the case study and analysis, data were taken from a solar photo-voltaic energy source linked to the Kigali national grid in the Rwanda bus system. The Electrical Transient Analyzer Program (ETAP 19) software is utilized for modeling and reliability analysis.

Keywords: distributed generation; energy resources; reliability assessment; Kigali national grid; ETAP

(Submitted on December 19, 2023; Revised on January 14, 2024; Accepted on February 19, 2024)

1. Introduction

Nowadays, the electricity grid is undergoing a favorable transformation due to both technological improvements and advancements regarding production about electricity-producing strategies [1]. The extensive reliance on coal and fossil fuels for power generation presents an immense threat to the planet, contributing significantly to climate change and global warming. As energy consumption escalates and environmental pollution intensifies, the urgency to seek alternative energy sources grows. Power system researchers are actively exploring integrating renewable energy resources (RER) into the electrical grid as a solution. These RERs represent green, eco-friendly energy sources that substantially benefit the environment [2].

Fuel prices are fast rising because of the limitations of fossil fuels, and various challenges such as worldwide warming, climate change, and harmful emissions have fueled demand for the implementation of RER [3]. DG multiple growing various multiple sources such as solar power, wind power, wave energy, biomass, and collaboration with the utility grid is rapidly growing via various sources such as solar power, wind power, wave energy, biomass, and cooperation with the utility grid. DGs, a subset of power systems, function as self-contained units directly linked to the consumer side, catering to specific client demands [4]. These compact power generators are engineered to supply low voltages directly to consumers. DG microgrids (MGs) are being incorporated into the electricity system to boost their efficacy and interoperability [5]. It to restructure a distribution network by decreasing its load in various areas, especially protecting devices, security, and dependability [6]. It has

* Corresponding author.
E-mail address: muyizeredarius@gmail.com

transformed the power distribution network in rural or distant locations by bringing electricity to those sectors through DG or loads. It offers grid alternatives and facilitates the utilization of DG and RER [7].

It provides several options for isolated places and regional power delivery. MG's essential characteristics that improve system efficiency are peer-to-peer power transmission and bilateral interaction. This point-to-point standard connection (PCC) operates as an interface that links the MG and the more fantastic more excellent utility network. Whenever there is a disruption in the primary utility grid, MG provides uninterrupted power delivery through PCC [8, 9]. It may be used in both electrically linked and island modes. These approaches generate energy through the use of DG [10].

The essential requirements of the customer are uninterrupted, secure, and dependable. Numerous interruptions and malfunctions in the equipment recently highlighted the need for enhanced resilience in the distribution system (DS). Aside from these challenges, technological breakdowns negatively hurt system dependability [11]. Table 1 demonstrates the reasons for distribution network disruptions. DS should remain the Rwanda Energy Group company's primary emphasis because it supports the final customer [12]. To optimize system efficiency, it is critical to evaluate and enhance the reliability of the electrical power system (DS), a factor responsible for as much as 97% of reliability problems affecting Rwanda's national grid [13-15]. The reliability of electricity systems can be evaluated with the following approach: The primary method is the analytical technique, which requires a series of approximation computations involving mathematics in the case of a complex system like the Kigali national grid. ETAP is used in this work to analyze and evaluate the DS.

Table 1. Summary of Annual Outages Occurred in the FY 2021/2022 [11]

Cause	Frequency	Duration	Energy not served (MWh)	Total cost	Freq %
Earth fault	3584	496.7	795.5	147,976,197	33%
Over-current	3551	487.8	677.2	125,959,778	32%
Under frequency	3327	326.4	463.5	86,202,751	30%
Emergency works	319	155.6	179.6	33,405,486	3%
Planned works	146	235.0	358.3	66,52,660	1%
Overload	59	24.3	64.6	12,020,498	1%
Overvoltage	7	0.4	4.7	877,623	0%
Grand Total	10,993	1,726.1	2,543.5	473,094,992	100%

This paper is structured as follows: Section 2 presents the reliability index analysis of load flow. Section 3 describes the methodology and proposed model. Section 4 presents the reliability index-based PV system. Section 5 displays the results of the discussions, and at the end, Section 6 presents the conclusion and future work of the paper.

2. Reliability Metric

Reliability indices assess the mathematical computation of reliability data for various networks of distribution elements, loads, and users. With this level of reliability, indications are typical values that show the overall dependability of a system [16]. Interruptions may cause either short-term or long-term drops in service to customers' voltages. They are sometimes referred to as dependability issues. Interruption is typically defined as a momentary or continuous interruption in interpersonal electricity. They are usually described as a significant issue. Persistent interruptions can also be characterized as disruptions that continue for over five minutes.

The IEEE developed an array of indices for assessing the reliability of distribution networks. The data above are divided into load point indices and electricity indices. Load point and energy indices are significant elements in reliable testing [17], and the mathematical modeling of several relevant indices is discussed here. Equations (1)-(2) assess the rate of system interruptions, whereas mathematical Equations (3)-(4) assess their period, electricity, and cost factors. These factors are critical in load forecasting, energy cost prediction, and load center outage control [18-20].

Reliability Indices for Reliability Design

The System Average Interruption Frequency Index (SAIFI) quantifies the frequency of sustained interruptions experienced by an average customer within a defined timeframe, typically spanning one year.

$$SAIFI = \frac{\sum_i C_i}{C_T} \tag{1}$$

C_i represents the proportion of affected clients for each incident, while C_T denotes the total number of consumers for which the index is computed.

The System Average Interruption Duration Index (SAIDI) denotes the cumulative duration of interruptions encountered by the average customer over a specified timeframe.

$$SAIDI = \frac{\sum_i C_i T_i}{C_T} \tag{2}$$

T_i refers to the duration necessary to recover from each interruption, i.

The Customer Average Interruption Duration Index (CAIDI) signifies the average time taken to restore service to customers following an interruption.

$$CAIDI = \frac{\sum_i C_i \times T_i}{\sum_i C_i} = \frac{SAIDI}{SAIFI} \tag{3}$$

The Average Energy Not Supplied (AENS) index quantifies the typical amount of energy that remains unavailable to consumers over a specific duration.

$$AENS = \frac{\sum_i P_i \times T_i}{\sum_i C_i} = \frac{ENS}{\sum_i C_i} \tag{4}$$

P_i indicates the quantity of work typically lost during outages, while C_i: signifies the absence of energy delivery, represented by interruptions.

The Average Service Availability Index (ASAI), shown as Equations (5)-(6), signifies the percentage of the reporting period during which customers had uninterrupted access to electricity.

$$ASAI = 1 - \frac{\sum_i C_i \times T_i}{C_T \times T} \tag{5}$$

T represents the duration of time, typically spanning 8,760 hours in a standard year or 8,784 hours in a leap year.

$$ASAI = \frac{T - SAIDI}{T} \tag{6}$$

3. Methodology and Proposed Model

- Flow Chart

The national grid of Kigali is modeled as ETAP for the purpose of analyzing the events. Figure 1 illustrates a proposed framework for assessing the reliability of the Distribution System (DS).

- Case Study System

Figure 2 illustrates how the electric power condition of the Kigali national grid, which includes the Substation 11 electrical system [9] and a Jali PV power plant, was evaluated for this study. The Mount Kigali SS9 is connected to the PV generator via a 15 KV/110 KV step-up transformer. According to Figure 2, the system consists of six synchronized generators, each with a four-bank capacity of 4.5 MVA and FACTS. For the modeling, design, analysis of load flow, reliability assessment (RA), and different evaluations of power system characteristics, ETAP 19 is the modeling tool employed [21-24].

- Reliability Specifications for PV Farm

The Jali PV farm will feature 18396 panels with a planned capacity of roughly 5 MW. Then, using a 10 MVA step-up transformer, the inverters link the PV arrays to a 15 KV bus coupled to a 110 KV bus. A blocked schematic for a PV farm-to-grid connection using a DC AC inverter may be found in Figure 3. Table 2 shows that the chosen inverter features the Maximum Power Point Tracker (MPPT), allowing for the most excellent possible harvest of solar energy.

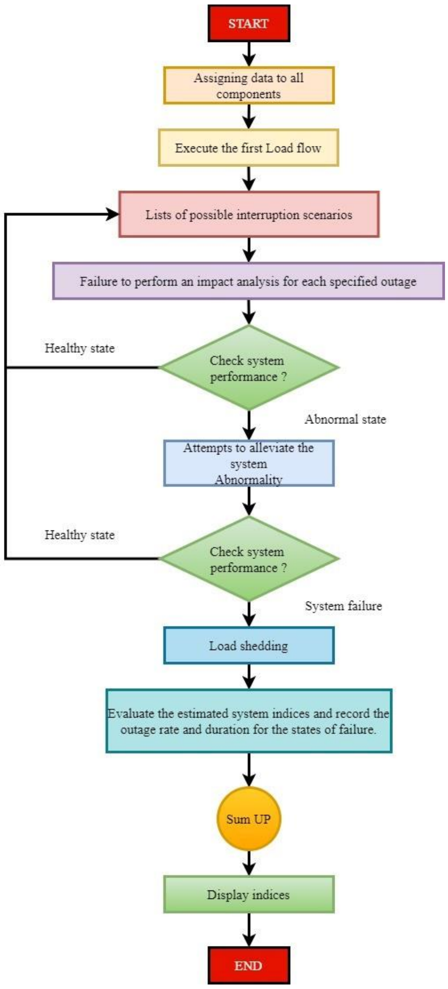

Figure 1. Application methodology for proposed work

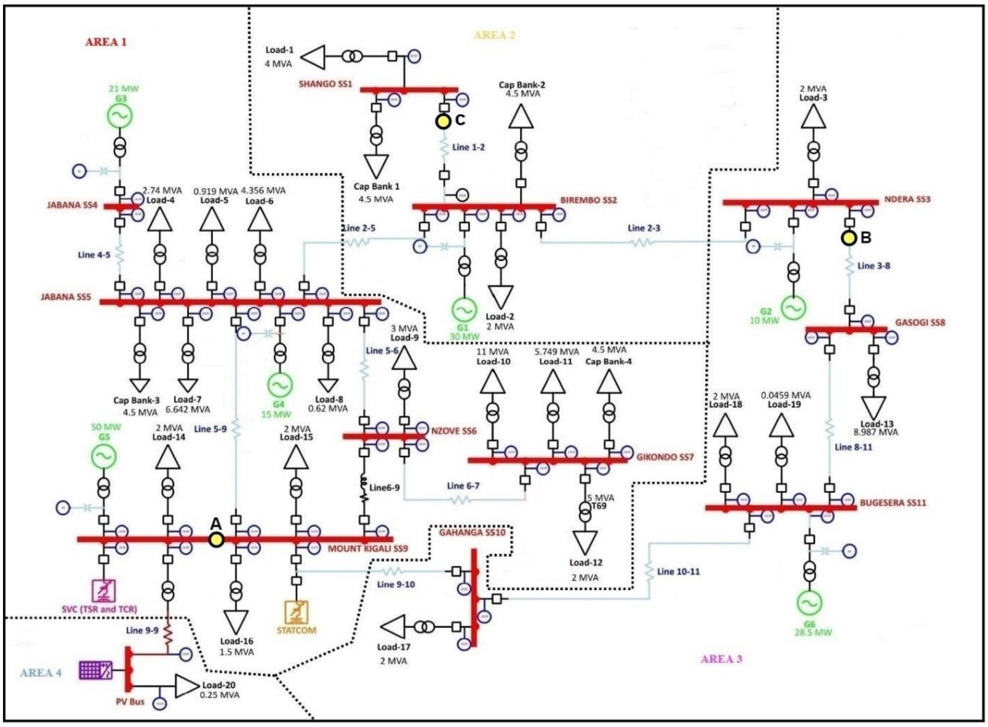

Figure 2. Reliability assessment of the Kigali national grid system using ETAP

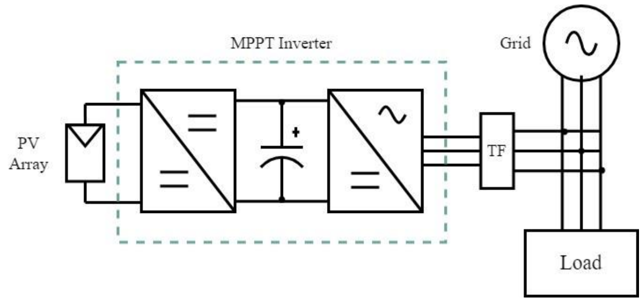

Figure 3. PV farm injecting both real and reactive powers

Table 2. Jali PV Farm Specification

Type of unit	Failure rate	Repair time	Switching time
5 MW	1.254 /yr	1752 /yr	5 hours

4. Case Study Results & Discussion

For RA assessment, three case studies were examined and assessed. In the initial case, reliability indices are evaluated without DG. Subsequently, the second case addresses the calculation of Energy Not Supplied (EENS) during outages, while the third case examines the impact of Distributed Generation (DG) on the distribution system.

- Case 1: Reliability assessment (RA) without DG

The ETAP-based framework design is depicted in Figure 3, enabling the evaluation of system indices through the RA tool. Figure 3 presents the components of the test system alongside reliability indices, showcasing the RA outcomes in scenarios where DG is absent at each load point. Table 3 displays these indices (SAIDI, SAIFI, and EENS), while ETAP 19 estimates a number of factors. These system settings display more significant numbers that follow IEEE specifications.

Table 3. Reliability indices without solar PV

System indexes	Without implementing solar PV
ACCI	3112.90 kVA /customer
AENS	99.4928 MW hr /customer. yr
ALII	0.83 pu (kVA)
ASAI	0.9968 pu
ASUI	0.00316 pu
CAIDI	36.646 hr /customer interruption
CTAID	27.646 hr/customer. yr
ECOST	0.00 $ /yr
EENS	1989.857 MW hr / yr
IEAR	0.000 $ / kW hr
SAIDI	27.6456 hr /customer. yr
SAIFI	0.7544 f /customer. yr

- Case 2: Estimating Energy Indices Across Load Points

The proposed model encompasses 20 loads distributed across different buses, vulnerable to disruptions from various sections or energy supply shortages to utility centers. These disturbances result in substantial financial losses for utility companies, translating into millions of dollars due to brief customer power shortages. The investigation initially considers the energy shortfall, quantified as Energy Not Delivered (EENS), treating it as the reference case. Table 4 details the impact of interruptions at these twenty loads on the system's reliability. Additionally, Figure 4 visually represents the customer distribution across these 20 load points within the network, their average consumption, and a graphical comparison of commonly utilized energy indices (specifically EENS), pivotal in assessing system reliability.

Table 4. Energy indices (EENS) estimation

Loads	Average interruption (f/year)	Outage Average Duration (hour)	Average interruption duration (hour/year)	EENS MWh (hour/year)
Load 1 [4.19MW]	0.9571	26.00	24.8827	104.2872
Load 2 [2.48MW]	0.3477	55.20	19.1945	47.4350
Load 3 [2.48MW]	0.5265	45.25	23.8219	59.1733

Load 4 [3.38MW]	0.6545	42.87	28.0570	95.0569
Load 5 [1.15MW]	0.3992	54.01	21.5590	24.7220
Load 6 [5.38MW]	0.4697	49.19	23.1044	124.4043
Load 7 [8.09MW]	0.6545	42.87	28.0570	227.0810
Load 8 [10.5MW]	0.6545	42.87	28.0570	21.6922
Load 9 [3.57MW]	0.9885	36.62	36.2035	129.3468
Load 10 [12MW]	2.1763	23.90	23.9000	634.9622
Load 11 [6.7MW]	2.0839	23.77	23.7700	330.9314
Load 12 [2.1MW]	2.2687	24.02	54.4964	111.0960
Load 13 [10MW]	1.1493	28.66	32.9365	345.6047
Load 14 [2.4MW]	1.1299	40.05	45.2532	112.1205
Load 15 [2.4MW]	2.6083	32.54	84.8743	206.0401
Load 16 [2.1MW]	1.7767	35.23	62.5875	127.5905
Load 17 [0.3MW]	0.8142	39.07	31.8074	9.9266
Load 18 [2.4MW]	0.1351	94.81	12.8086	30.7822
Load 19 [2.4MW]	1.3611	33.99	46.2588	110.3219
Load 20 [0.1MW]	2.9319	30.14	88.3562	4.9191

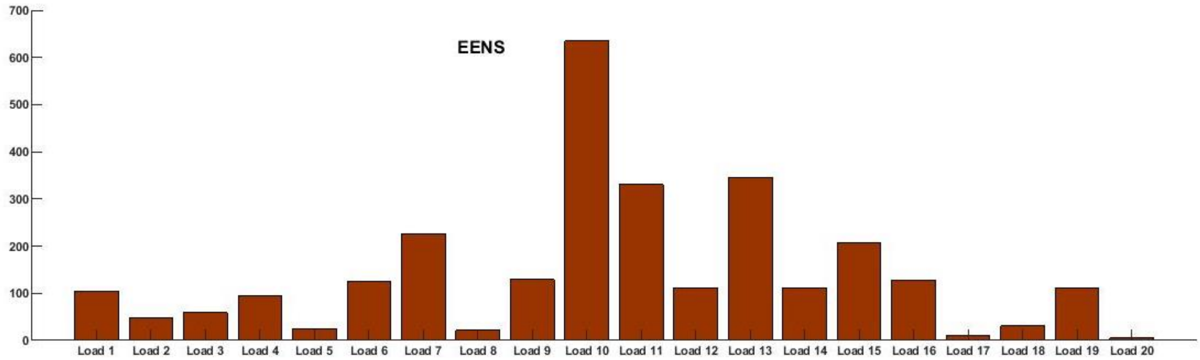

Figure 4. Comparison of Energy indices (EENS) of different 20 loads

- Case 3: Evaluating Reliability Post-Distributed Generation Implementation

The Mount Kigali SS9 is linked to a Jali PV with a 5MW rating as the DG. System indices are calculated utilizing the ETAP RA tool after the implementation. Table 5 presents the resulting indicators after the deployment of DG, showcasing reductions in values for SAIDI, SAIFI, ASUI, and EENS indices. Lower values in these indices signify an increase in reliability. Furthermore, the analysis regarding the distribution system's reliability with DG integration persists.

Table 5. Reliability indices with solar PV

System indexes	Implementing solar PV
ACCI	2830.37 kVA /customer
AENS	91.2499 MW hr /customer. yr
ALII	0.75 pu (kVA)
ASAI	0.9970 pu
ASUI	0.00301 pu
CAIDI	37.045 hr /customer interruption
CTAID	26.409 hr /customer. yr
ECOST	0.00 $ / yr
EENS	1824.999 MW hr / yr
IEAR	0.000 $ / kW hr
SAIDI	26.4094 hr /customer. yr
SAIFI	0.7129 customer. yr

- Comparison of Cases

Thus, the findings are highly comparable by comparing the estimated numbers with the simulated results from the reliability 440 analysis report performed using the ETAP program. This demonstrates the precision of our computations. These findings provide us with a greater knowledge of how the system performs regarding dependability, which is critical for power system control and planning. Furthermore, doing a thorough reliability study aids in the reduction of undesirable difficulties, outages, and blackouts, eventually leading to an increase in power quality. According to our findings, the failure rate in the projected distributed power system is shockingly high. It is critical to solve this issue to ensure a more dependable and efficient power supply.

Figure 5 depicts the base-case scenario devoid of distributed generation (DG) at individual load points, showcasing the system components and reliability indices analysis. The findings reveal that having DG solely at the endpoint doesn't affect the reliability near the substation. Yet, a comparison between system reliability with and without DG shows that integrating DG enhances overall reliability within the distribution system.

In Figure 5, the SAIDI, SAIFI, ASUI, and EENS indices illustrate a nonlinear relationship with average failure rates, restoration time, and average loads at individual load points. Analyzing the index variations compared to scenarios without PV, the SAIFI consistently shows a 5.50% decrease. SAIDI also notably drops by 4.47% compared to the base case, although CAIDI sees a slight increase of approximately 1.08%. Furthermore, Case 3 experiences an EENS decrease of about 8.28% compared to Base Case 1, yet ASAI demonstrates an increase of roughly 2.00% compared to Case 3 at solar PV Jali Farm.

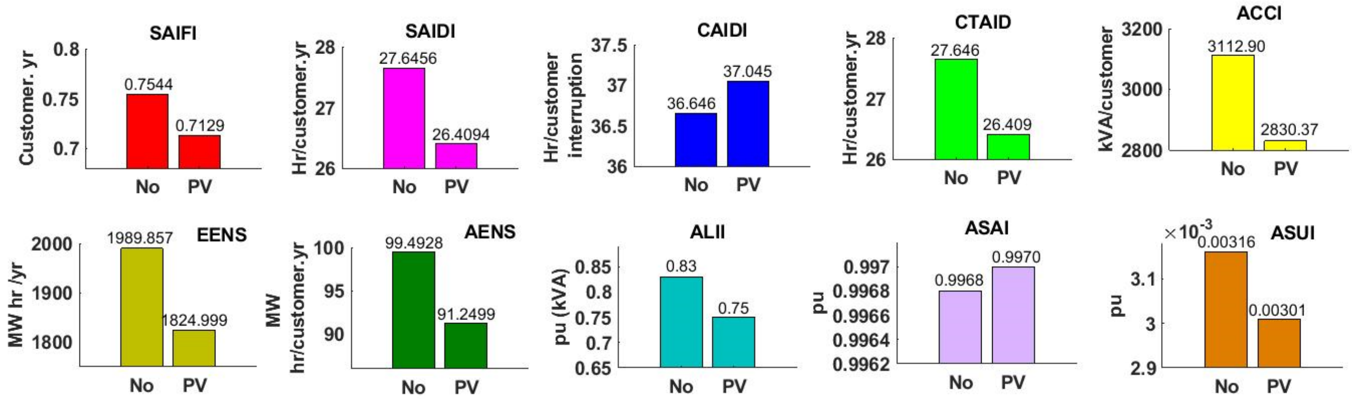

Figure 5. The base-case scenario devoid of distributed generation at individual load points

5. Conclusion

This paper centered on assessing the reliability of post-DG implementation within the Kigali national grid. The outcomes from three distinct case studies unveiled varied effects of DG integration on the distribution network's dependability. Table 5 illustrates an enhancement in system indices due to the inclusion of DG within the distribution system. Following DG deployment, reliability improved as SAIDI decreased by approximately 16%, SAIFI by 5%, EENS by 58%, ASUI by 5%, and ACCI by 9%. Introducing the ETAP-19 software for reliability assessment presents a novel method that mitigates human errors associated with past computational approaches. In conjunction with the suggested model, the software notably lowered the indices to levels recommended by IEEE standards, yielding more favorable outcomes. Future endeavors may involve using multiple DGs with diverse ratings and PV cells. Employing various methodologies could optimize DG placement to achieve maximum dependability. Additionally, conducting reliability assessments on various distribution systems, such as the IEEE 118-bus test system, using ETAP, remains an avenue for forthcoming research.

Acknowledgments

This work is supported by The African Centre of Excellence in Energy for Sustainable Development (ACE-ESD) at the University of Rwanda (Research Core Funding No. ACE II-017).

References

[1] Yan, T., Tang, W., Wang, Y., and Zhang, X. Reliability Assessment of a Multi-State Distribution System with Microgrids Based on an Accelerated Monte-Carlo Method. *IET Generation, Transmission & Distribution*, vol. 12, no. 13, pp. 3221-3229, 2018.
[2] Adefarati, T. and Bansal, R.C. Application of Renewable Energy Resources in a Microgrid Power System. *The Journal of Engineering*, vol. 2019, no. 18, pp. 5308-5313, 2019.
[3] Zhi-jian, L., Gui-hong, W., Dong-hui, Y., and Qi, S. Reliability Assessment of Distribution Network Considering the Randomness of Distributed Generation. In *2016 China International Conference on Electricity Distribution (CICED)*, IEEE, pp. 1-6, 2016.

[4] Karki, R., Billinton, R., and Verma, A.K. eds. *Reliability modeling and analysis of smart power systems*. Berlin: Springer, 2014.

[5] Cagnano, A., De Tuglie, E., and Mancarella, P. Microgrids: Overview and Guidelines for Practical Implementations and Operation. *Applied Energy*, vol. 258, pp. 114039, 2020.

[6] Lu, W., Ren, Z., Xu, J., and Chen, S. Edge Blockchain Assisted Lightweight Privacy-Preserving Data Aggregation for Smart Grid. *IEEE Transactions on Network and Service Management*, vol. 18, no. 2, pp. 1246-1259, 2021.

[7] Ansari, S., Chandel, A., and Tariq, M. Enhanced Power Sharing of Droop Controlled Differently Rated Inverters in Standalone AC Microgrid. In *2021 Fourth International Conference on Electrical, Computer and Communication Technologies (ICECCT)*, IEEE, pp. 1-6, 2021.

[8] Palati, M., Prathap, S.S., and Nagaraju, N.H. Modeling and Analysis of Smart Power System. *Artificial Intelligence-based Smart Power Systems*, pp. 15-36, 2023.

[9] Alam, M.N., Chakrabarti, S., and Ghosh, A. Networked Microgrids: State-of-The-Art and Future Perspectives. *IEEE Transactions on Industrial Informatics*, vol. 15, no. 3, pp. 1238-1250, 2018.

[10] Ansari, O.A., Safari, N., and Chung, C.Y. Reliability Assessment of Microgrid with Renewable Generation and Prioritized Loads. In *2016 IEEE Green Energy and Systems Conference (IGSEC)*, IEEE, pp. 1-6, 2016.

[11] Zhu, Y. and Singh, C. Topology-Aware Reliability Assessment by Graph Neural Networks. In *2022 IEEE Kansas Power and Energy Conference (KPEC)*, IEEE, pp. 1-6, 2022.

[12] Karngala, A.K. and Singh, C. Reliability Assessment Framework for the Distribution System Including Distributed Energy Resources. *IEEE Transactions on Sustainable Energy*, vol. 12, no. 3, pp. 1539-1548, 2021.

[13] Maraaba, L., Almuhaini, M., Habli, M., and Khalid, M. Neural Networks Based Dynamic Load Modeling for Power System Reliability Assessment. *Sustainability*, vol. 15, no. 6, pp. 5403, 2023.

[14] Chandraratne, C., Naayagi, R.T., and Logenthiran, T. Smart Grid Protection through Self-Healing. In *2017 IEEE Innovative Smart Grid Technologies-Asia (ISGT-Asia)*, IEEE, pp. 1-6, 2017.

[15] Rwanda Energy Group, Annual Report for Rwanda Energy Group 2020-2021, https://www.reg.rw/fileadmin/REG_ANNUAL_REPORT_2020-2021_V3.pdf, access on February 1, 2024.

[16] Ahmed, K.M.U., Alvarez, M., and Bollen, M.H. A Novel Reliability Index to Assess the Computational Resource Adequacy in Data Centers. *IEEE Access*, vol. 9, pp. 54530-54541, 2021.

[17] He, M., Liu, Y., Liu, H., and Gu, D. A Novel High Step-Up/Step-Down Transformer-Less Bidirectional DC-DC Converter for Energy Storage Systems. In *2021 IEEE/IAS Industrial and Commercial Power System Asia (I&CPS Asia)*, IEEE, pp. 1409-1414, 2021.

[18] TAŞIN, A. and ÇAVUŞ, T.F. Reliability Analysis of Sakarya Area Electricity Distribution System. *Sakarya University Journal of Science*, vol. 26, no. 2, pp. 292-299, 2022.

[19] Huda, A.S.N. and Živanović, R. Distribution System Reliability Assessment using Sequential Multilevel Monte Carlo Method. In *2016 IEEE Innovative Smart Grid Technologies-Asia (ISGT-Asia)*, IEEE, pp. 867-872, 2016.

[20] Mamo, A. and Hizkiel, A. Reliability Assessment and Enhancement of Dangila Distribution System with Distribution Generation. *Cogent Engineering*, vol. 10, no. 1, pp. 2191375, 2023.

[21] Saeed, M.H., Fangzong, W., Kalwar, B.A., and Iqbal, S. A Review on Microgrids' Challenges & Perspectives. *IEEE Access*, vol. 9, pp. 166502-166517, 2021.

[22] Aderibigbe, M.A., Adoghe, A.U., Agbetuyi, F., and Airoboman, A.E. A Review on Optimal Placement of Distributed Generators for Reliability Improvement on Distribution Network. *2021 IEEE PES/IAS PowerAfrica*, pp. 1-5, 2021.

[23] Muyizere, D., Letting, L.K., and Munyazikwiye, B.B. Decreasing the Negative Impact of Time Delays on Electricity Due to Performance Improvement in the Rwanda National Grid. *Electronics*, vol. 11, no. 19, pp. 3114, 2022.

[24] Mansoury, I., El Bourakadi, D., Yahyaouy, A., and Boumhidi, J. A Novel Decision-Making Approach Based on a Decision Tree for Micro-Grid Energy Management. *Indonesian Journal of Electrical Engineering and Computer Science*, vol. 30, no. 2, pp. 1150-1158, 2023.